

data-model

Data Model — Workshop Curriculum Platform

Feature: specs/001-workshop-curriculum-platform | **Date:** 2026-09-01
Derived from: [spec.md](#) Key Entities, constrained by [research.md](#)

Scope, and what is authoritative

This document is a deliberate abstraction. It is NOT the full schema, and a field's absence here is not evidence that the field does not exist. It carries the entities, their relationships, the identity decision they all hang on, and the state machines. Storage columns, wire fields, DDL, closed enums and gate definitions live in the contracts, which are normative:

Layer	Authoritative document
Registry records, SQLite DDL, ingest matching, resolution, redaction log	contracts/passage-contract.md
HTTP request/response bodies, status envelopes, all closed enums	contracts/http-api.md
CLI surface, exit codes, pipeline stages	contracts/pipeline-cli.md

Where this document and a contract disagree, **the contract wins and this document is the defect**. Section [Not modelled here](#) names the specific fields this abstraction omits, so a reader is pointed rather than left to infer absence.

The one decision everything hangs on

Passage.pid is a **ULID minted once at ingest**. It is not derived from content and not derived from position.

This is not a style preference. Both alternatives were measured and both fail (D-SEARCH-1):

Candidate identity	Behaviour measured	Consequence
Lumen chunk ID (content-derived)	typo fix changed 4b98e295c112a429 → 610e608a50f273a6	every human correction (FR-038) silently breaks every citation to that passage
Positional key (file + line range)	prepending a section moved every later range	every edit above a passage breaks its links

Under either, **SC-016 scores 0% on corrected passages while the UI keeps rendering links that now point somewhere else** — a failure with no visible symptom. A minted identifier is the only form that survives both operations, which is why it is the root of this model.

Entities

Chapter

Field	Type	Notes
ordinal	int	zero-padded on disk (chapters/01/), matching existing convention
slug	string	stable, URL-safe; never reused
title	string	human title
summary	string	shown in the chapter list (FR-008)
recording	Recording	

Field	Type	Notes
		exactly one today; the model permits several
materials	[]SupportingMaterial	notes, diagrams, code
transcript	Transcript	absent until produced
status	enum	draft transcribed published

Rules: a chapter MUST NOT reach published with an incomplete material set (FR-028). `ordinal` and `slug` are immutable once published — cross-references resolve through them.

Recording

Field	Type	Notes
path	string	git-ignored; the whole file is never committed
parts	[]string	the 36×50 MB <code>tar.gz</code> parts that <i>are</i> committed
sha256	string	verified before use (FR-007)
duration_s	float	measured, not assumed — Chapter 1 is 6928.75
audio	AudioProps	codec, sample rate, channel layout

Rules: reassembly MUST verify `sha256` and fail loudly on mismatch. `duration_s` is the denominator for SC-001 coverage, so it is stored, not recomputed per query.

`AudioProps` records `dual_mono`: `true` for Chapter 1 — measured L–R difference –90.3 dB. This is data, not trivia: it is why `Passage.speaker` cannot be machine-derived (D-TRANS-2).

Transcript

Two layers, because the machine output is *evidence* and must survive correction (FR-038).

Field	Type	Notes
chapter_slug	string	owner
engine	string	e.g. faster-whisper/large-v3-turbo
engine_params	map	including the frozen normaliser hash
machine_passages	[] <code>Passage</code>	immutable once written
corrections	[] <code>Correction</code>	append-only overlay
accuracy	<code>AccuracyReport</code>	absent until measured

Rules: `machine_passages` is never mutated — a correction is a new `Correction` row. Deleting it would destroy the ability to re-measure accuracy later (SC-002).

Passage — the addressable unit

Field	Type	Notes
pid	ULID	minted at ingest ; the only citable identity
kind	enum	transcript doc_section code diagram
source_ref	<code>SourceRef</code>	where it came from; may move without changing pid
text	string	current text: machine, or corrected if a <code>Correction</code> applies
provenance	enum	machine human_corrected (FR-038)
confidence	float?	avg_logprob-derived; null for non-transcript kinds

Field	Type	Notes
uncertain	bool	drives the FR-003 “unclear, not guessed” marking
t_start_s, t_end_s	float?	transcript kinds only; the SC-003 5 s bound applies
speaker	string?	human-attributed or null — never machine-guessed (D-TRANS-2)
redacted	bool	FR-039; suppresses from index, answers and export
content_hash	string	change detection only — deliberately NOT identity

Rules: - pid is written back into the source artifact as `<!-- pid: ... -->` so the anchor travels with the text. - For a chapter’s transcript passages: `U [t_start_s, t_end_s] U VAD_silence == [0, duration_s]` **exactly**. This makes SC-001 an arithmetic identity rather than a judgement (D-TRANS-4). - content_hash is named to prevent the exact mistake this model exists to avoid. Any code treating it as identity is a defect. - A redacted passage **MUST** propagate to the index and to any stored answers, not merely to the rendered transcript.

Correction

Field	Type
pid	ULID (the passage corrected)
text	string (replacement)
author, at, reason	provenance

Rule: corrections never change pid. That is the whole point of the layer.

CrossReference

Field	Type	Notes
from_pid, to_pid	ULID	directed

Field	Type	Notes
relation	enum	explains demonstrates continues related
origin	enum	derived authored
score	float?	for derived

Rules: self-references rejected. Traversal must be cycle-safe (spec edge case: “cross- references must not create loops that trap a reader”). derived links are rebuildable; authored links are content and survive re-index.

Index (derived, never authoritative)

Two paths, because one cannot meet both latency targets (D-SEARCH-2).

Path	Backing	Measured
lexical	SQLite FTS5 prefix index over symbols/ titles	p95 9.58 ms
semantic	embedding vectors, 768-dim	query embed 18.2–21.0 s under load

Field	Type	Notes
generation	int	monotonic
state	enum	building verified live superseded discarded (passage-contract.md §4.4 generations)
pid_count	int	verification gate input

Rules: only a verified generation may become live, by atomic swap. Readers never see building — measured necessity: during the live rebuild, chunks moved while `last_indexed_at` still advertised the previous generation, i.e. a half-written readable index.

Answer

Authoritative schema: [contracts/http-api.md](#) §3.10 and §5. The table below is the entity view of that contract. It uses the contract's field names and the contract's enum members exactly — see [Arbitration](#) for why.

Field	Type	Notes
question	string	
status	enum	answered declined unavailable — the three terminal states. (pending is a state of the answer <i>job</i> resource, not of an answer; http-api.md §3.10.)
text	string?	present only when answered
citations	[]ULID	≥1 when answered , enforced by schema (minItems: 1)
reason	discriminated	absent when answered; shape and vocabulary determined by status — table below
retrieval	RetrievalTrace	top score and margin, recorded even on success

reason is discriminated by status, and the two reason vocabularies are disjoint:

status	HTTP	reason shape	Closed vocabulary
answered	200	absent	—
declined	200 — a correct result, not an error	bare enum string	below_threshold margin_too_small unsupported no_citations redacted_evidence (http-api.md §5.5)

status	HTTP	reason shape	Closed vocabulary
unavailable	503	object { code, message, retry_after_s }	the answering-leg codes of http-api.md §5.3: no_provider provider_disabled provider_unreachable model_not_generative verification_unavailable

Rules: answered with zero citations is structurally undecodable (minItems: 1). Every citation MUST be a member of the live generation’s PID set (generation_members) — a deterministic check, and the reason SC-009 is reachable at all. A citation outside the set makes the whole answer declined{no_citations}; a citation that resolves to a redacted passage makes it declined{redacted_evidence}; a citation whose verification *could not run* makes it unavailable{verification_unavailable}, never declined — verification that could not run is not verification that failed (passage-contract.md §7.2 C3–C5). Claims are never silently stripped. retrieval is stored even on success so a 0.002-margin pass is visible as *fragile* rather than indistinguishable from a confident one.

unavailable is a first-class third state, distinct from declined. Conflating them would make a missing provider look like a content gap — the Honest Instruments principle applied to the domain.

Why no_provider is not a decline reason. A decline is a *judgement about the content*: the system retrieved, evaluated, and determined the corpus does not support an answer. unavailable is the absence of the instrument that would have made that judgement. “No generative model is installed” is not a fact about the curriculum, and filing it as a refusal reason would report a thing that could not run as a thing that was judged — the exact failure the three-state contract exists to prevent. no_provider therefore lives only in the unavailable vocabulary (http-api.md §5.3), and the two vocabularies above share no member.

Arbitration (2026-09-01) — one Answer shape

This entity previously carried a second, incompatible shape: verdict / refused / refusal_reason, with no_provider filed among the refusal reasons. That fork is **withdrawn, not silently replaced**, and the record is kept here so a reader of an older draft can see what changed:

	withdrawn (this document)	adopted (http-api.md)
field name	verdict	status
refusal member	refused	declined
reason field	refusal_reason	reason
		below_threshold, margin_too_small, unsupported, no_citations, redacted_evidence —
reason members	below_threshold, margin_too_small, unsupported, no_provider	and no_provider moved to the unavailable vocabulary, where its meaning belongs

Three reasons the API contract's shape won:

1. **It is the shape the tests will be written against.** T057–T068 cite `http-api.md`; no task cites this document for the Answer shape. `plan.md` marks refusal behaviour and citation verification [TDD], so those tests land *before* the implementation — against whichever document they name.
2. **verdict was already taken.** `submodules/verdict/pkg/verdict/` is this project's 0/1/2 three-valued *instrument* result, used by every command and endpoint (T018). Naming the answer's domain outcome `verdict` too would put two different meanings on one word in one codebase, which is the conflation this feature is built to avoid.

Two corrections here, both 2026-09-01, both path/reference only — the naming argument itself is untouched. `pkg/verdict/` was an unqualified path written when the package lived under the backend module; `verdict` is now a separately published module mounted at `submodules/verdict` (see [tasks.md](#)'s File Structure correction). ~~(T012)~~ was simply the wrong task number — T012 is the append-only redaction log; the verdict helpers are **T018**. Found while correcting the paths, and fixed by editing the *citation*, never by renumbering the task: other documents and agent reports cite these T-numbers, so moving one to make a stale reference true would break every correct reference to it.

3. **status is already the discriminator everywhere else.** The search envelope keys on status (§2.5), and declined is the term already used by passage-contract.md §7.2 (C3–C5), research/llm-bridging.md (“Behaviour when the provider is unreachable”) and the SC-010 gate. Choosing status/declined moved four documents into agreement instead of two.

Relationships

Chapter 1—1 Recording

| 1—0..1 Transcript —1—* Passage —0..* Correction
| 1—* SupportingMaterial —1—* Passage



Passage *—* Passage (CrossReference, directed, cycle-safe)

Passage *—1 Index generation (membership; rebuildable)

Answer *—* Passage (citations; membership checked against the LIVE generation)

State transitions

Chapter: draft → transcribed → published. Regression to draft is permitted (a recording is replaced); published → published must be idempotent (FR-027).

Index generation: building → verified → live, and live → superseded on swap. building → discarded on failed verification — a generation that fails its gate is never served, which is what keeps FR-020 honest.

Passage provenance: machine → human_corrected only. There is no path back, because the machine text is retained separately rather than overwritten.

Validation rules traced to requirements

Rule	Requirement
pid minted, never derived	FR-037, SC-016
machine passages immutable; corrections layered	FR-038, SC-002
span union equals full duration	FR-001, SC-001
timestamp within 5 s	FR-002, SC-003
	FR-003

Rule	Requirement
uncertain set from engine confidence, never inferred	
speaker null unless human- attributed	FR-005
sha256 verified before reassembly	FR-007
redacted propagates to index and stored answers	FR-039
answered $\Rightarrow \geq 1$ citation, all in live PID set	FR-021, SC-009
declined $\neq$ unavailable; their reason vocabularies are disjoint	FR-022, FR-025, SC-010, SC-013
code passage identity keyed on (path, symbol) + symbol_aliases; unaliased rename fails loudly	FR-037, SC-016a
only verified generations serve	FR-020
re-ingest is idempotent on pid	FR-027

Not modelled here — and where it does live

The contracts specify fields this document deliberately does not repeat. They are **named** here so that absence above reads as *abstraction*, not as *nonexistence*. Duplicating them would create a fourth place to drift, which is precisely how the Answer fork above happened.

Field / table	What it carries	Authoritative location
machine_text	the immutable machine layer that makes SC-002 re- measurable	passage-contract.md §4.2 (F1, F2), §4.3 DDL
source_ref.anchor	inline sidecar registry_only — how the identifier travels	passage-contract.md §4.2, §8 R5/R6
uncertain_reason	low_confidence inaudible overlapping_speech	passage-contract.md §4.2 F3; http-api.md §3.4

Field / table	What it carries	Authoritative location
speaker_source	language_uncertain; non-null iff uncertain human unattributed — never machine (D-TRANS-2). This is the FR-005 “state where it cannot” half.	passage-contract.md §4.2 F4; http-api.md §3.4
symbol_aliases	the code-passage rename alias table SC-016a rests on	passage-contract.md §4.3 DDL, §6.2–6.6, §8 R3/R4
redactions	append-only redaction log — Passage.redacted above is the <i>materialisation</i> , not the record	passage-contract.md §4.3 DDL, §7.3
generation_members	the pid set citation verification checks membership against	passage-contract.md §4.3 DDL, §7.2 C2
sections[]	FR-011 navigable chapter structure (id, title, first_pid, t_start_s)	http-api.md §3.3
content_hash, minted_at, ingest_run	change detection and identifier provenance	passage-contract.md §4.2

Deliberately out of the model

Unlike the section above, these have **no** authoritative location — they are absent by decision.

- **User accounts and permissions** — decision D1 makes this single-user and local.
- **Direct audio/video embeddings** — Lumen cannot index media (D-SEARCH-3); media is reached through transcripts. Stated rather than implied.
- **Distributed or multi-writer state** — single machine, single writer.